



CorpusMind Voice

Offline audio to linguistically annotated corpus · A companion tool for CorpusMind

SPEECH TO CORPUS. NO CLOUD.

USER GUIDE

VERSION 1.3.0

100% OFFLINE

ENGLISH + العربية

CorpusMind Voice turns recordings (MP3, MP4, M4A, AAC, OGG, OPUS, WAV, FLAC, WEBM, WMA, AMR, 3GP, AIFF) or live microphone input into a linguistically annotated corpus on your own machine, then analyses it academically. No account, no upload, no cloud API. The app embeds its own Whisper inference engine and ffmpeg decoder, so real transcription never depends on an internet connection or an installed Python stack.

1 Install and set up

- a Web:** run `bun install`, `bun run db:push`, `bun run dev`, then open `http://localhost:3000`. Installable as a PWA for on-the-go recording, entirely on-device.
- b Desktop:** install the Windows (NSIS/MSI), macOS (Apple Silicon/Intel dmg) or Linux (deb) build from the GitHub release. The Whisper engine and ffmpeg ship inside the installer.
- c First tab: Settings & Diagnostics.** Download a Whisper model (`tiny`, 75 MB, for tests; `large-v3-turbo`, 900 MB, for research accuracy). Models persist in the app data folder and are reused fully offline. Since v1.3.0 all recordings live in the per-user app data folder, writable in every install mode.

2 The six-stage pipeline (Studio)

Drop a file, browse, or record from the microphone. Pick the **language** (English, Arabic: Egyptian **عامية**, Arabic: MSA **فصحى**), the **device** and the **Whisper model**. Processing starts automatically:

#	Stage	What happens
1	Ingest	container decoded to 16 kHz mono PCM by the bundled ffmpeg
2	ASR	local Whisper (INT8/Q4) with word timestamps from cross-attention DTW
3	Alignment	word boundaries aligned to the waveform from Whisper attention heads
4	Prosody	Praat-style F0 (50-400 Hz), intensity, jitter, shimmer, HNR measured on the audio
5	Disfluency	pauses > 200 ms, fillers (<code>um</code> , <code>uh</code> , <code>آه</code> , <code>يعني</code>), repeats, false starts, lengthenings
6	Structure	SQLite (3 tables) + JSON + TEI/XML written locally

3 Correct the transcript

High ≥ 0.85

Mid 0.60-0.85

Low < 0.60

In the **Transcript Editor** every token carries its confidence as a colour; skim the red tokens first and click any token to correct it (the utterance text is rebuilt so exports always match). Each utterance lists its disfluencies and prosody summary, with **click-to-play** for the span, **speaker relabeling**, and an **editable filler lexicon** in Settings.

4 Re-run or delete

Every entry in the Studio's **Recent jobs** list carries two buttons.

Re-run picks a different Whisper model, dialect or device and reruns the whole pipeline: test with `tiny`, finish with `large-v3-turbo`. The pipeline card also offers Re-run as soon as a job finishes or fails. **Delete** removes the audio with its transcript, analysis and metadata, after confirmation.

5 Linguistic analysis (eight modules, computed locally)

- **Overview** - tokens, types, speech rate, hapax, confidence distribution.
- **Frequency** - counts, per-1,000 rates, Gries **DP dispersion**, Zipf rank-frequency curve.
- **KWIC** - context window, case toggle, timestamps; match modes: normalized / whole-word / regex.
- **Keywords** - G^2 , **LogRatio** and **%DIFF** effect sizes (Hardie 2014).
- **Collocates** - node-word explorer (span 1-5) with **MI**, **t-score**, **logDice**.
- **Lexical diversity** - MATTR (window 50), MTLD, TTR, lexical density.
- **Readability** - Flesch / Flesch-Kincaid (English), language-neutral LIX.
- **Speech & Prosody** - disfluency rates per 1,000 tokens, pause statistics, prosody aggregates.

6 Corpus Overview & metadata

Metadata describes the session (title, speaker, dialect, date, place, genre, licence); fields are embedded into the TEI `<teiHeader>` of every export. The **Corpus Overview** tab aggregates every session on the device: inventory, corpus frequency with DP dispersion, and cross-session KWIC with playback. The full UI switches to right-to-left Arabic from the العربية toggle.

7 Save to device

- **JSON** - pandas-ready records
- **CSV** - flat token table
- **TEI/XML** - header + aligned text
- **SQLite** - three-table corpus database

Plus **Praat TextGrid**, **ELAN EAF**, **SRT** and **WebVTT** transcript exports; every analysis table saves to CSV and the report to JSON.

8 Local AI assistant (optional)

The **Assistant** tab chats with a local model through **Ollama** (<http://127.0.0.1:11434>) or **LM Studio** (<http://127.0.0.1:1234>), and offers three per-transcript tools: **Summarise**, **Preview without fillers**, and **Suggest topic tags**. A custom **Ollama host** (e.g. 192.168.1.50:11434) can be saved in Settings to use a daemon on another machine. If no provider is reachable, the panel says so and never substitutes a cloud service.

9 Phone companion (optional, desktop)

In **Settings & Diagnostics** → **Phone companion**, click **Enable**, restart the app once, then scan the **QR code** with the phone (same Wi-Fi). The phone gets the same app behind a pairing token: record or upload, transcribe with the desktop engine, browse, export and chat. Use the <https://> link for phone-side **microphone recording** (browsers allow the mic only on secure pages; accept the self-signed certificate once). Nothing leaves the local network, and disabling the card closes the service.

10 Troubleshooting

Job failed: model not downloaded	Settings & Diagnostics → Whisper models → Download , then Re-run the job
Job failed: decoding error	unusual container codec; convert to WAV/MP3 and re-upload
Microphone button does nothing	grant mic permission in browser/OS, or upload a file; recording needs HTTPS or localhost
Upload/recording fails with a permission error (v1.2.x)	fixed in v1.3.0: data moved to the per-user app data folder; update the app, older recordings migrate automatically
Upload/recording fails with "Internal Server Error" (installer built 2026-09-11)	packaging defect in the first v1.3.0 installers: the speech runtime failed to load in the installed app. Re-download the installer (rebuilt 2026-09-12); the web/PWA was never affected
Phone cannot open the pairing link	same Wi-Fi, restart the app after enabling (card shows Active), allow the app on private networks in the firewall
Assistant unreachable	start <code>ollama serve</code> or load a model in LM Studio (port 1234); for a remote daemon save its host in Settings
Downloaded model "vanished"	it cannot; models persist in the app data folder shown in Settings